

Name: _____

Student ID: _____

Problem 3 [10 pts] Pass or fail

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} \mid x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} \mid x = 4)$?

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

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Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

- (d) Sketch $P(\text{pass} \mid x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.

